

## Project 1 — Classification of Abnormal Electric-Vehicle Charging Sessions

### 1. Context

Electric-vehicle charging infrastructures generate large volumes of operational data. Identifying charging sessions at increased risk of **abnormal termination** can support preventive monitoring, charger maintenance, resource allocation, and service-quality improvement.

In this project, you will develop and evaluate a classification solution for predicting whether an electric-vehicle charging session will terminate abnormally.

Each group must design, justify, implement, and critically evaluate its own modelling solution.

### 2. Prediction objective

The objective is to predict the binary target:

- is\_Abnormal = 1: abnormal charging session;
- is\_Abnormal = 0: normal charging session.

The prediction must represent a decision made **when a charging session starts**. Consequently, only information that would reasonably be available at that time may be used as a predictor.

The project should answer the following general question:

To what extent can contextual, operational, and historical information available at the beginning of a charging session be used to **predict abnormal termination of a charging session**?

### 3. Available data

The dataset provided for this project is **an educational adaptation** of the China EV Charging Dataset introduced by Zhang et al. (2025). The original dataset contains 441,077 charging transactions recorded over two years at 13 public charging stations located in three districts of Jiaxing, China.

The project dataset retains variables describing the complete charging transaction, including information recorded at the beginning, during, and after the charging session. It also includes electricity-tariff information and precomputed historical attributes describing previous activity and abnormal sessions associated with charging posts and users.

The available variables are organised into the following groups.

#### Session identification and location

- UserID: identifier of the user associated with the transaction;
- Charging Post ID: identifier of the charging post;

- Location Information: charging-station location;
- District Name: district in which the station is located.

#### **Charging-transaction information**

- Order creation time: date and time of the order charge
- Transaction power/kWh: energy associated with the charging transaction;
- Electricity cost/Yuan: cost of the electricity consumed;
- Service charge/Yuan: service fee applied to the transaction;
- Transaction Amount/Yuan: total transaction amount;
- Actual Payment/Yuan: amount effectively paid;
- Start Time: date and time at which the charging session started;
- End Time: date and time at which the charging session ended;
- Payment time: date and time at which payment was recorded;
- End cause: reason why the charging session ended.

#### **Weather conditions**

- Temperature(°C): recorded temperature;
- Relative Humidity(%): recorded relative humidity;
- Precipitation(mm): recorded precipitation.

Consider weather variables as estimates or forecasts available at the beginning of the charging session.

#### **Electricity-tariff information**

- electricity\_price: electricity price applicable at the session start time;
- tariff\_period: tariff interval determined by the session start time.

The dataset contains precomputed attributes describing previous activity at the same charging post and at the same user, including:

#### **Historical charging-post information**

- numbers of previous completed sessions over different temporal windows;
- numbers and smoothed rates of previous completed abnormal sessions;
- time since the previous completed session;
- time since the previous completed abnormal session;
- indicators identifying posts without previous completed-session or abnormal-session history.

#### **Historical user/account information**

- is new user
- numbers of previous completed sessions over different temporal windows;
- numbers and smoothed rates of previous completed abnormal sessions;
- time since the previous completed session;
- time since the previous completed abnormal session;
- indicators identifying new users/accounts and accounts without completed-session history;
- number of other sessions associated with the same account that were active when the current session started.

Earlier-started sessions that remained active at the prediction time were excluded from all outcome-derived historical attributes. Their outcomes were not yet available. Concurrent activity is represented separately through `user_active_sessions_at_start`, which uses only start and end timestamps available at the prediction time.

Historical attributes were constructed chronologically to prevent the current session or future sessions from contributing information to the current predictors. Students are not required to reconstruct these variables, but they must:

- understand their meaning;
- evaluate their contribution through feature-group ablation;
- consider whether they would be operationally available at prediction time;
- discuss their implications for privacy, generalisation, and cold-start cases.

#### 4. Main tasks

##### Task 1 — Understand, audit, and prepare the data

Conduct a systematic and critical analysis of the supplied data before developing the classification models. The objective is to understand the dataset, identify factors that may affect modelling, and define a justified data-preparation strategy.

As guidance, the analysis may include aspects such as data types and variable meanings, missing values, duplicated or inconsistent observations, temporal coverage, target-class distribution, numerical distributions, potential outliers, categorical cardinality, rare categories, relationships between variables, temporal changes, and possible data-quality problems. These examples are not exhaustive.

Potential data-quality problems must be investigated before corrective action is taken. Decisions to retain, transform, impute, group, cap, or remove observations must be supported by evidence and by the meaning of the corresponding variables.

Explore the data using appropriate numerical summaries and graphical representations. Figures should be selected to answer relevant analytical questions and must be interpreted in the report. A collection of plots without an explanation of their meaning, modelling implications, or relationship to the project objective will not be considered an adequate exploratory analysis.

The exploratory analysis should consider whether important properties of the data change over time. In particular, investigate whether session volume, class prevalence, predictor distributions, user or charging-post activity, and other relevant patterns remain stable throughout the observation period. Explain how any identified temporal changes may affect model development and evaluation.

Any preparation operation that learns information from the data—including imputation, scaling, category grouping, outlier thresholds, resampling, feature selection, or dimensionality reduction represent different methodological reasons and must be documented separately.

At the end of this task, provide a concise synthesis identifying:

- the most important characteristics and problems discovered in the data;
- the evidence supporting the adopted preparation decisions;
- the variables or variable groups retained for modelling;
- unresolved limitations or assumptions;
- the expected implications for model selection and evaluation.

The quality of this task will be assessed through the relevance and depth of the investigation, the justification of decisions, and the interpretation of evidence—not by the number of statistics, figures, or preprocessing operations presented.

Each weekly submission should include the corresponding notebook and a concise report section explaining decisions, evidence, results, and conclusions—not merely describing the code.

The report should focus on reasoning, evidence, and critical analysis. It should not reproduce sections of source code.

## References

Zhang, Y., Xu, T., Chen, T., et al. (2025). A high-resolution electric vehicle charging transaction dataset with multidimensional features in China. *Scientific Data*, 12, 643. <https://doi.org/10.1038/s41597-025-04982-1>